Low-Cost Automation of Multi-Property Characterisation Spatial Mapping for Composition Spread Alloy Films (CSAF) using Open-Source CNC3018 Platform

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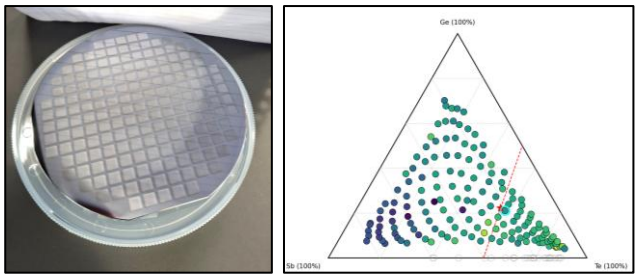
Abstract: High-throughput characterisation of Composition Spread Alloy Films (CSAF) is critical for accelerating Phase Change Materials research and development such as Phase Change Memory (PCM). However, most commercial automated instruments are financially prohibitive. This paper presents a low-cost and modular automation system developed on a repurposed CNC3018 mechanical platform, to map the electrical and optical properties of CSAF. Featuring a modular swappable probe head, the system performs automated four-point probe (4PP) resistivity measurements and spectral reflectance mapping. The system was validated using precision resistors (10 Ω to 10 kΩ) and materials such as 12µm Aluminium foil, which yield a mean bulk resistivity ρ of $2.99 \times 10^6\ \Omega\ cm$, comparable to a pure Al baseline $2.65 \times 10^6\ \Omega\ cm$. Validation reveals good precision from low coefficient of variation (CV%) and fulfilment of the 3-sigma criteria. This paper demonstrates the measurement of a Germanium-Antimony-Tellurium (GST) CSAF wafer, an alloy widely used for PCM, during its amorphous state (~25°C) and after annealed to nominal 200°C, and 300° C. The system successfully identified stoichiometric GST-225 alloys, and we mapped the bulk resistivity transitions at $\rho = 331.48\ \Omega\ cm$ (amorphous) to $\rho = 0.0548\ \Omega\ cm$ (200°C crystalline) and $\rho = 0.0154\ \Omega\ cm$ (300°C crystalline), alongside its change of optical reflectance characteristics for multiple wavelengths. These results demonstrate a highly reproducible, multi-property automation platform that makes high-throughput materials research more accessible.

Keywords: *Composition Spread Alloy Films, Automated Spatial Mapping, Low-Cost Scientific Instrumentation, Four-Point Probe (4PP), Reflectance Spectrometry, Thin-Film Characterization, Resistivity, Reflectance, Phase Change Memory (PCM).*

I. INTRODUCTION

Composition Spread Alloy Films (CSAF), or can be referred to as *Combinatorial Spread Alloy Films*, are a library consisting of many different alloy chemical compositions, synthesised as thin films on a single substrate [1]. Current materials research commonly work with CSAF as they are essential in high-throughput materials research to rapidly test new different material compositions, including phase change alloys. To enable quicker data analyses of compositions and decision making, researchers create characteristic spatial and ternary maps of the wafers [2]. Spatial maps show the composition of alloy as located on the wafer, while a ternary map show the density of the element across a triangular graph that make up the entire wafer. Each corner of the ternary map represents the maximum percentage of each element.

In recent times, emerging applications of phase change alloys, such as Phase Change Memory (PCM), is already being researched extensively as a response to the growing demand of non-volatile memory to replace flash technology. These phase change alloys are designed to undergo reversible transitions between highly resistive amorphous and highly conductive crystalline states to store logical '0' and '1', resulting in structural changes that shift both electrical resistivity and optical reflectance. The most widely used alloy for PCM application is Germanium-Antimony-Tellurium (abbr. $GeSbTe$ or GST), a chalcogenide compound capable of rapid switching at high speeds between its amorphous and crystalline states via thermal excitation. For its role in PCM research, its characteristics have been studied extensively across different alloy compositions such as $GeTeSb_2Te_3$ and $Ge_2Sb_2Te_5$ [3], [4]. The following **Fig.1** shows an example of a GST sample and its composition map.



(a) (b)

Fig 1. (a) a wafer containing $GeSbTe$ composition film, (b) a composition ternary map for Gb, Se, Te.

However, performing electrical and optical characteristic measurements requires a high entry cost to even procure commercial instruments. Commercially available manual electrical measurement instruments such as four-point probe systems can easily cost thousands of dollars, which include specialised probe heads and specialised probe stands. Automated measurement systems allow for high-throughput data collection and analysis which greatly accelerates materials discovery [2], but is even more prohibitively expensive, which means that cost alone limits the accessibility for academia, researchers with budget constraints, and even small industries. While researchers have attempted to create low cost solutions such as using homemade four-point probe equipment [5] or modifying open-source hardware (such as 3D printers) for making automated spatial maps [6], these existing solutions typically focus on single-purpose characterization (either electrical-only or optical-only scans). If both properties must be mapped, two separate systems would be required.

To address these financial and technical challenges, this paper aims to develop a low-cost, modular, and autonomous wafer spatial mapping system using an open-source 3-axis CNC machine as the actuator.

Presented in the following **Table 1** are the detailed Requirement criteria and descriptions for a Minimum Viable Product (MVP):

	Requirement	***Goal***	***Minimum Viable Product***
1	Scanning Dimension	100 mm diameter (4-inch) wafers	CNC3018 (300x180mm bed), custom Tray to hold 100 mm wafers or 10x10cm square sample
2	Material Type	Composition Spread Alloy Films (CSAF)	Optimized for characterization of Ge-Sb-Te (GST) film
3	Application	Phase Change Memory (PCM)	Thermal transition analysis of GST films across three states: Amorphous (25℃), annealed with hot plate to 200℃ and 300℃
4	Target Properties	Spatial mapping of electrical/optical properties	Automated scanning with Python program to control the CNC and receive sensor telemetry
5	Precision	3-sigma rule (99.7% data within 3 std deviations from mean)	Rigorous validation using precision resistors and prepared samples (Al, Graphite).
6	Scanning Time	Complete a full scan in 1 hour.	Full 177-point scan completed in ~30 minutes
7	Data Density	at least 177 points (entire CSAF library on wafer)	177 points for resistivity measurement, 225 points (15x15) for spectral measurement across a 10cm square.
8	Electrical Characterization	Sheet resistance (Rs) and bulk resistivity (ρ)	Dual mode circuit: Low Resistivity (10 Ω to 1 kΩ/sq) and High-Resistivity (>1 kΩ to GΩ/sq), with Picoammeter and MCP6002 Op-Amps
9	Optical Characterization	Reflectance (R) in common wavelengths	AS7343 spectrometer capturing visible and NIR bands (405 nm to 855 nm).
10	Variations	Amorphous (25°C), Crystalline (200°C), Crystalline (300°C)	Capable of testing properties after hot plate crystallisation annealing
11	Modular Probe Architecture	Modular probe heads for electrical and optical measurements	Swappable bracket mount: A: DIY inline 4-Point Probe (needle-tip pogo-pins) B: Commercial Jandel Macor Professional Probe C: SparkFun AS7343 non-contact spectrometer
12	Benchmarking Baselines	Verify against established baselines/ literature	Benchmarked against commercial reference standards, prior research, and measuring equipment (Digital Multimeter)

Table 1. Requirements and Description of a Minimum Viable Product

The rest of the paper is structured as following:

a) Section II details the theory of Electrical and Optical measurements of thin-film within this paper, including 4-Point Probe Measurement, Transmission Line Method (TLM) and calculation of optical Reflectance.
b) Section III details the hardware design and specifications to create the MVP as described in Table 1.
c) Section IV describes the algorithm and design used in the electrical and optical characteristic mapping software. Section V presents the validation strategies and results of the system to fulfil the MVP requirements,
d) Section VI presents the analysis of results for measurement on a GST CSAF sample, as well as describing the transition of its electrical and optical characteristics across Amorphous and Crystalline states.
e) Section VIII presents the conclusion and future work.

II. MEASURING THIN FILM ELECTRICAL AND OPTICAL PROPERTIES

1. Resistivity with 4-Point Probe Method

Electrical characteristics describe how a material will behave when connected to electricity. While there have been many methods of electrical characterisation, the 4-point probe method for measuring semiconductor sheet resistance and bulk resistivity has become widely used in the semiconductor industry as standard for monitoring production processes [7], as it allows measurement of the sheet resistance or resistivity of semiconductor materials and thin films while eliminating lead and contact resistance errors.

For comparison, if we use a multimeter with two leads, a 2-point measurement of a semiconductor in **Fig. 2(a)** would result in the value of total resistance (R_T), defined as

$$R_T = \left(\frac{V}{I}\right) + 2Rc \qquad \textbf{(Eq.1)}$$

When we perform a 4-point probe measurement in **Fig. 2(b)**, a direct current (I) is applied through the two outer probes, and the voltage drop (V) is measured across the two inner probes. This configuration decouples the current-driving path from the voltage-sensing path, and eliminates Rc. Hence,

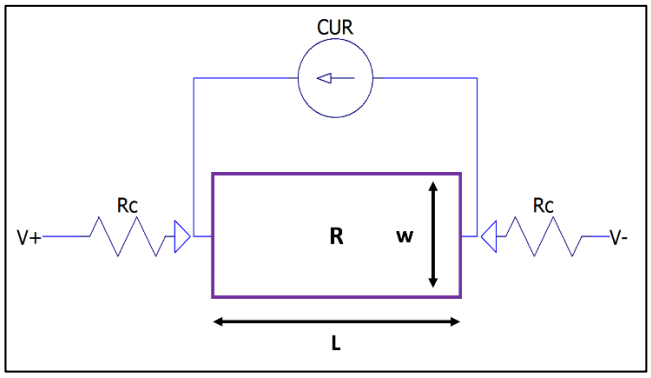
$$R = \frac{V}{I} \qquad \textbf{(Eq.2)}$$

4-point probing also allows us to find the sheet resistance of an infinitely large thin sheet by multiplying **(Eq.2)** by a geometric correction factor of the radial 2D current flow from 4 evenly spaced contact points.

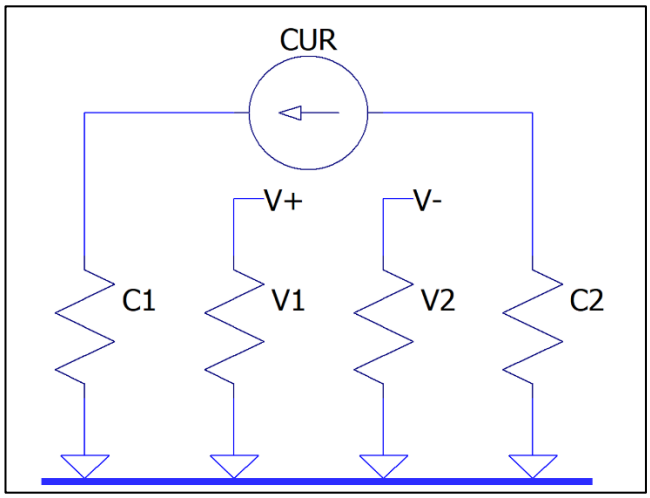
$$R_s = \frac{\pi}{\ln(2)}\left(\frac{V}{I}\right) \approx 4.5324\left(\frac{V}{I}\right) \qquad \textbf{(Eq.3)}$$

Finally, the bulk resistivity of the material is the sheet resistivity multiplied by its thickness (t):

$$\rho = R_s * t \qquad \textbf{(Eq.4)}$$



(a)



(b)

Fig 2. (a) 2-point probe measurement (b) 4-point probe measurement

2. *Resistivity with Transmission Line Method (TLM)*

Another method of measuring sheet resistance and resistivity of a material is to with a series of 2-point measurements across the length of a material with uniform width [8]. As this method reveals the effects of lead and contact resistances, we can use this method as validation of the four-point probe measurement.

A semiconductor in **Fig.4** with width W and length L has the resistance:

$$R = Rs * \frac{L}{W} \qquad \textbf{(Eq.5)}$$

Hence, its total resistance R_T in **(Eq.1)** can be rewritten as:

$$R_T = Rs * \frac{L}{W} + 2Rc \qquad \textbf{(Eq.6)}$$

As we can see, R_T has linear relationship with L, similar to the slope-intercept formula of a straight line:

$$y = mx + b \qquad \textbf{(Eq.7)}$$

Sheet resistance (Rs) can then be measured using linear regression as the slope of a series of 2-point measurements times the width of the semiconductor.

$$Rs = m * W \qquad \textbf{(Eq.8)}$$

3. *Optical Reflectivity and Reflectance*

A material has intrinsic characteristic which is its ability to reflect light and other forms of energy from its surface at ideal conditions, which is the optical reflectivity (measured in ratio of reflected energy to the total energy)**.** This depends on the refractive index of the material and the angle of incidence, or the angle of which light strikes the material. [9]

The intrinsic reflectivity percentage of a material at a certain wavelength can be calculated from the Fresnel Equation for reflectivity at normal incidence (R):

$$R = \left(\frac{n1 - n2}{n1 + n2}\right)^2 \qquad \textbf{(Eq.9)}$$

$n1$: Refractive index of the initial medium (e.g. 1.0 for air).
$n2$: Refractive index of the second medium (e.g.1.5 for glass).

However, CSAF wafer is a kind of inhomogeneous thin film [10] that has non-uniform refractive index, since its chemical composition changes along the different compositions on the wafer, and it also has varying thickness. Hence a slightly different approach is taken to measure its ability to reflect light, which is Optical Reflectance.

Optical reflectance is the relative measure of a material's ability to reflect light and other forms of energy from its surface, regardless if its refractive index is not known. Like reflectivity, It is defined as the ratio of reflected energy to the total energy striking the object [11]. Since the energy of a white light consists of different wavelengths, an optical spectrometer is used to measure the different values of reflectance for a given wavelength. Reflectance percentage is defined as:

$$R(\lambda) = \frac{I_{sample}(\lambda) - I_{dark}(\lambda)}{I_{reference}(\lambda) - I_{dark}(\lambda)} \qquad \textbf{(Eq.10)}$$

$I_{sample}(\lambda)$: The measured intensity of the sample
$I_{reference}(\lambda)$: The reference intensity at max reflectance
$I_{dark}(\lambda)$: The reference dark intensity at min reflectance

The intensity values are obtained via experiment to calibrate the reflectance percentage calculation. It is necessary to subject the sensor to a completely reflective surface (such as a white piece of copy paper) and a completely dark surface, to get the threshold to separate ambient light from the actual measured energy of the light source.

III. HARDWARE DESIGN

1. *System Architecture*

The hardware is divided into three subsystems:

a) The automated mechanical actuator for positioning the sensing probe of targeted property (electrical/optical). Custom 3D-printed brackets will mount the sensors to the moving head of the actuator. Several upgrades such as adding limit switches for enabling homing and a 3D printed sample tray are implemented.

b) The Resistivity measurement system with 4-point probe, the appropriate probe holder, the data acquisition circuit using an Arduino microcontroller, an ADC module, and several benchtop laboratory equipment including a Power Supply Unit (GWInstek PPE3223) with Voltage and Current Control and a Pico Ammeter (Keithley 485) for detecting low current in High Resistance mode.

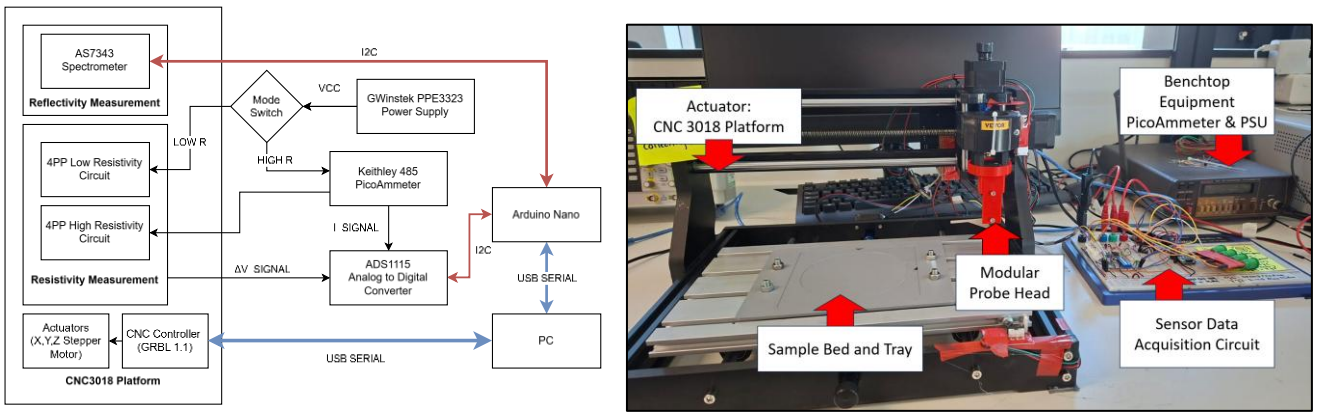
c) The Reflectance measurement system with an optical spectrometer (AS7343), a spectrometer sensor bracket, and the data acquisition circuit using Arduino. We also need a white paper as a white calibration sample, black cloth as a dark calibration sample, and a cloth shroud to cover the whole system while performing the measurement to ensure no ambient light comes in.

The following **Table 2** describes the specifications and how the components integrate for each subsystem, and **Fig.3** shows the system architecture and actual implementation.

Subsystem	*Component*	*Source*	*Specification*	*Integration*
(a) Actuator	CNC3018 Platform	COTS	3-Axis movement frame with high resolution stepping	USB Serial to PC
	Microswitches	COTS	Limit switches on XYZ axes to enable homing and position tracking	Wired to CNC interrupt pins
	Custom Bed Tray	Custom Design, 3D Printed	Custom tray for 100mm wafer (0.4mm recess) or 10x10cm square sample	Mounted to CNC bed (Al. Extrusion 15180) with M6 bolt and nuts
	Sensor Brackets	Custom Design, 3D Printed	Swappable mounting bracket for DIY 4PP, Jandel Macor 4PP, and AS7343 Spectrometer	Mounted to CNC 42mm spindle slot
Telemetry MCU	Arduino Nano	COTS	Microcontroller for Sensor Telemetry (ATmega328P)	USB Serial to PC
(b) Resistivity Measurement	DFRobot ADS1115 ADC Breakout Board	COTS	16-bit High Resolution ADC for measuring 4PP voltage drop and Picoammeter analogue output	I2C Protocol to Arduino
	Op Amp MCP6002 (1)	Custom Circuit	5V compatible op-Amp, 10T**Ω** input Impedance for High-Resistivity Circuit ADC input	Wired between 4PP Voltage probes to ADS1115 input
	Op Amp MCP6002 (2)	Custom Circuit	Reverse mode, Flip voltage polarity of Picoammeter output	Wiring between Picoamp Output to ADS1115 input
	DIP Switch Box	Custom Circuit	Selector for High (5V 2mA) or Low Resistance Mode (5V 100mA)	Wiring between 4PP sensor and telemetry system
	Keithley 485 Picoammeter	COTS (Lab Equipment)	Sensing small current (range: 2nA to 2mA) for High-Resistivity Circuit, output is analogue -1V to 1V	Wired in series with the main current loop, disabled on High Resistivity mode
	GWInstek PPE3223	COTS (Lab Equipment)	Programmable DC power supply with Voltage Control and Current Control (set to 5V)	4PP Current Source
(c) Reflectance Measurement	SparkFun AS7343 Spectrometer	COTS	14-channel visible light /NIR spectrometer (405nm - 855nm)	I2C Protocol to Arduino Nano

Note: COTS = Commercial Off-The-Shelf

Table 2. Specification and Description of Subsystem



(a) (b)

Fig 3. (a) High Level Architecture Diagram, (b) Photo of System Implementation

2. *CNC3018 platform*

This system is built on a repurposed consumer 3-axis CNC3018 machine. The machine uses three NEMA17 stepper motors on each axis, with a theoretical step resolution of 1.25 µm (800 steps/mm). There is a general assumption regarding stepper motor applications, the real-world accuracy is subject to an error tolerance of approximately $\pm 5\%$ to the desired step position or angle.

An unmodified CNC3018 would have to be manually positioned by the operator to a starting position due to the control system being an open-loop control. To enable an absolute and reproducible coordinate system, we add physical limit switches, which were wired into the control board interrupt pins to allow "homing", a command for the machine to move to its absolute minimum position across the three axes, and automatically set a zero-machine position or 0,0,0.

The sample is affixed to the CNC bed using a custom 3D printed tray that will hold square 10cm samples or a standard 4-inch (100mm) wafer. The tray has a 0.4mm deep circle to hold the wafer in place. Probe holder brackets for DIY 4PP sensor, Jandel Macor 4PP sensor, and AS7343 Spectrometer are also designed to be compatible with the CNC's 42mm spindle hole and 3D printed using PLA Filament **(Fig.4)**.

3. *Resistivity Measurement Circuit*

The system is designed be able to perform 4-point probe measurements on a wide range of resistivity values. Measuring low resistivity data and high resistivity data require different approaches in its hardware design.

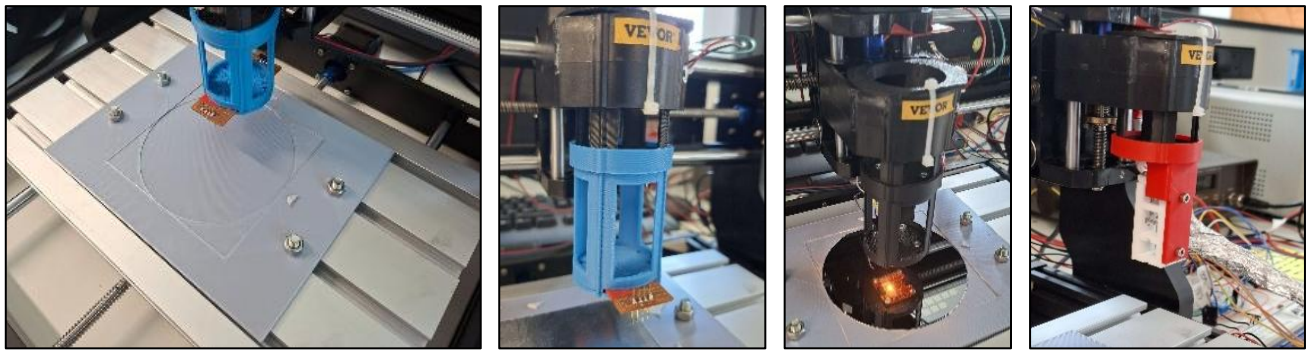
a. *Low Resistivity Measurement (Fig. 5(a))*

A low resistance sample such as aluminium has a theoretical bulk resistivity of 2.65 to 2.82 µΩ cm at room temperature [12]. Performing measurement on such material would result in very tiny voltage drop across the inner voltage probes. It requires higher injected source current and a finer resolution (smaller range) for the voltage sensing. Having a current more than 2mA is impossible to measure using the Keithley 485, hence this circuit bypasses the Pico Ammeter entirely, and since there is low impedance across the circuit, an op-amp is not needed to get clean data in the ADC.
Using a 5V 100mA source, the sensing circuitry is relatively simple, and relies on the high resolution of the ADC.

b. *High Resistivity Measurement (Fig. 5(b))*

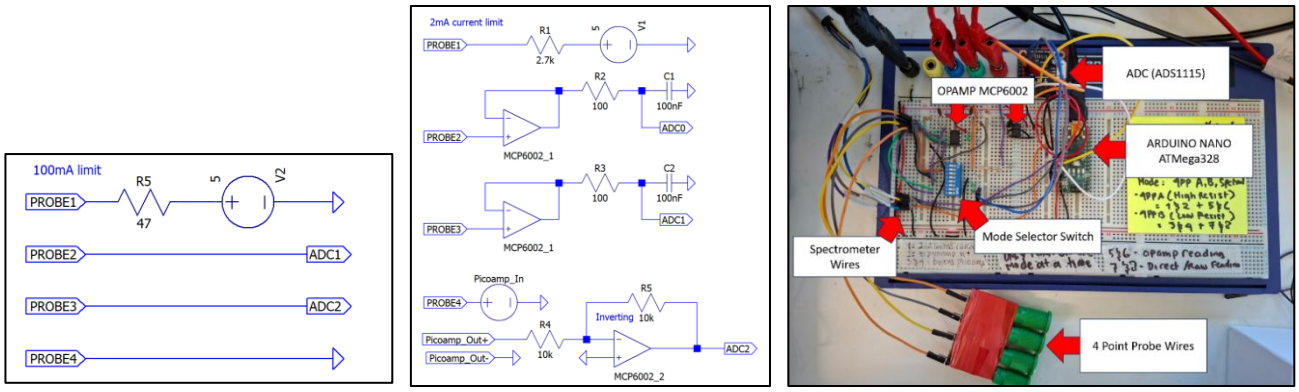
However, a high resistance sample such as the phase-change alloy $Ge_2Sb_2Te_5$ (GST225), in its amorphous state has the bulk resistivity value of 28 kΩ cm at 25°C [4]. With thickness between the range of 100 to 200 nm, the sheet resistance would be around 1.4~2.8 GΩ/sq. Performing measurement on this sample needs a lower current (nanoampere-microampere range) and therefore requires an additional Pico Ammeter which is a current measurement device designed to sense very small currents (nA range).

Furthermore, sensing voltage drop requires an ADC with a wide voltage dynamic range and a high input impedance, higher than the resistance of the sample, to eliminate circuit loading errors across the inner probes. This is solved using the MCP6002 OpAmp with ~10T**Ω** input Impedance.



(a) *(b)* *(c)* *(d)*

Fig 4. (a) Custom Sample tray affixed to the CNC bed, (b) Custom DIY 4PP bracket, (c) Custom Jandel Macor Probe bracket, (d) Custom AS7343 Spectrometer bracket



(a) *(b)* *(c)*

Fig 5. (a) Low Resistivity Mode Circuit, (b) High Resistivity Mode Circuit, (c) Photo of Implementation on Breadboard

To select between the Low and High Resistivity circuits, a physical box switch is implemented in the circuit **(Fig, 5(c))** in which the user can select the appropriate switches to enable one of the circuits at a time (in this case, 1-2-5-6 is the High Resistivity Mode, and 3-4-7-8 is the Low Resistivity Mode). This setup can be replaced with a digitally controlled switch setup as a further improvement.

As for the probes, we have made a DIY inline four-point probe using equally spaced Needle-tip Pogo Pins soldered to perfboard **(Fig. 4(b)).** This probe is used during the design validation stage (Chapter V) and on materials other than thin-film wafer, to avoid damaging the Jandel Macor Probe. The Jandel probe is to be used only for the thin-film wafer samples.

4. Optical Reflectance Measurement

The spectrometer used is the low-cost SparkFun AS7343 Breakout Board (Sensor: AS7343, 14-channel multi-spectral, I2C interface). This sensor is selected due to its cost effectiveness and high number of channels for its price (capability to pick up 14 different wavelengths), and the simplicity as its breakout board allows communication with the Arduino via I2C Protocol.

It has a built in LED as the light source for measurement mounted right beside the sensor, therefore we are able to assume by pointing the spectrometer straight down using the onboard LED, and perform reflectance measurement at normal incidence (0°) **(Fig.4(c))** The LED is covered with one layer of masking tape to ensure a more diffused light.

IV. SOFTWARE DESIGN

This section describes the algorithm and design used in the electrical and optical characteristic mapping software.

1. High-Level Architecture

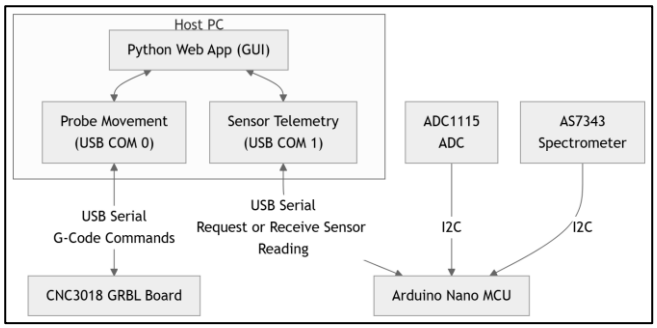
This project's software design objective is to develop a Graphical User Interface (GUI) on top of a software to automate motion of the CNC platform, coordinate tracking, and serial data acquisition to obtain 4-point probe data and spectrometer data. By automating the probe's movements in the CNC platform, the user will not have to manually enter G-code strings for each point of measurement.

The custom software system was developed in Python. The software functions as a centralized coordinator on a PC. The PC will send the G-codes over a dedicated USB serial port to the CNC control board and acquiring synchronized physical measurements in real time. At the same time, another USB serial port of the PC communicates with an Arduino Nano, which streams parsed, high-resolution sensor telemetry captured by the ADC (for Resistivity Measurement) or Spectrometer (for Reflectance Measurement).

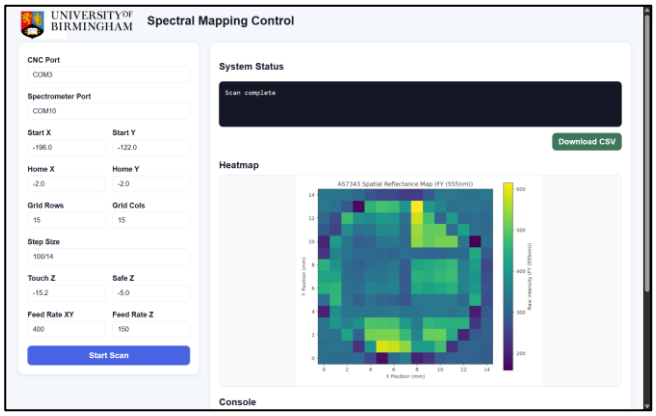
2. User Interface Design

The User Interface was developed using a Python and Flask web application framework. The decision to use a web app format was to create cross-platform versatility, enabling the system console can be accessed from any host device with a web browser without requiring platform-specific installations or environments.

The system's high level software architecture and a screenshot of the user interface is presented in **Fig.6.**



(a)



(b)

Fig 6. (a) High-level Software Architecture, (b) Screenshot of the User Interface

3. Algorithm Design

The system's overall algorithm is as following:

1. Start Scan: The system initiates either resistivity or reflectance automated using parameters defined by user (Start Position, Grid Size, Z Depth)
2. Serial Initialisation: The program initialises serial communication to the CNC controller and the MCU.
3. Machine Homing: hardware homing cycle to establish absolute machine zero coordinates.
4. Path Planning: The system calculates the target XY grid coordinates and traversal order across the sample area.
5. XY Positioning: The gantry navigates to the target node's horizontal X and Y coordinates.
6. Z-Axis Lowering: The Z-axis lowers the sensor to the target height.
7. Send Read Command Trigger: The host transmits the trigger command to request telemetry data.
8. Telemetry: The MCU performs a reading and streams back raw data.
9. Data Logging: The host software logs the parsed telemetry data and record the current node position
10. Loop: The loop checks whether additional unmeasured targets remain in the scheduled path.
11. Return Home: The Z-axis safely retracts and the gantry moves back to the home clearance position.
12. Data Export: The collected data is written to a CSV file and generates the final 2D spatial heatmap plot.
13. System Idle: The scan routine closes active hardware serial ports and returns the system to an idle state.

The complete system flowchart alongside link to maintained repository is presented in **Appendix 1.**

V. DESIGN VALIDATION

This chapter presents the steps taken in performing the experiments and validation of both the electrical (resistivity) and optical (reflectance) measurement using the hardware and software designs presented in the previous two chapters.

The two main metrics in this validation process are:

1. Standard Deviation (σ): It is the measure of variation of repeated measurements [13]. A lower σ means that there is low variation of the data and therefore higher precision. In the case of measurement devices, it describes how much the measurements fluctuate, which lower is better.

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{n}} \qquad \text{(Eq.10)}$$

x_i: measurement of point
μ: mean of the samples
n: number of samples

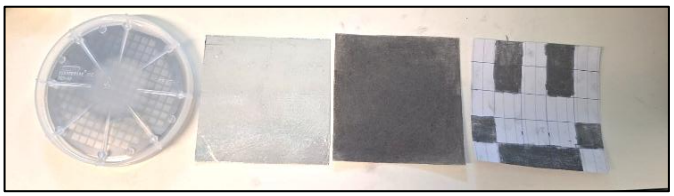
A standard practice to determine good precision is the 3-sigma (3σ), a manufacturing standard of precision that has also been used in mass metrology [14], which states that 99.7% of all data points in a normally distributed process must lie within 3 times the standard deviation. Within this, 95% must fall between two sigmas, and 68% within one sigma.

2. Coefficient of Variation (CV): It is the spread of variation around the mean, which represent the system's stability when the variation scales with the magnitude or measurement scale of the mean [15].

For example, the standard deviation of resistivity in the Ω vs in k Ω will have different magnitude scales. Expressing the variation in CV means that no matter the difference of magnitude of the data, we can compare their relative variations.

$$CV = \frac{\sigma}{\mu} * 100\% \qquad \text{(Eq.11)}$$

The samples that are used for validation areas below (Fig.7):



Left to right:

a. GST CSAF Sample (S705, provided by Microsoft). It is measured at 25°C (Amorphous), and crystallised annealed states at 200°C and 300°C
b. Aluminium Foil, assumed to be 12 microns thick
c. Pencil Graphite (6B) on copy paper, assumed ~6 microns thick layer
d. Pencil Graphite on paper forming a smiley face.

Fig 7. Prepared Validation Samples

For the resistivity measurement setup, testing on precision components (such as precision resistors), allows us to find the general precision of the ADC before using 4-point probing. To avoid damage to a provided GST Composition Spread Film wafer and the Jandel 4-point probe, the design is tested on other materials first (precision resistor, aluminium, and pencil graphite on paper) before scanning on GST film.

1. Electrical Resistivity

a) Measuring Known Precision Resistors

To test the general accuracy of the Low and High Resistivity measurement circuits (Fig. 5(a-b)) using certified known resistance values, measurement is performed with precision resistors replacing the 4PP in the circuit. The resulting measurement results are compared with their rated values and values as read by Digital Multimeter (Rapid 318 DMM) as presented in **Table 3**. We can see the normal distribution curves of the precision resistor measurements confirm that while the standard deviation increases predictably at higher resistance values, the system has considerably high relative precision across four different resistor values, with all points satisfying the 3 sigma and 2 sigma rules. Points in 1 sigma between 60-80% suggesting precise values. The difference compared to DMM values are between ~0.4 to 4.1%. Our measurements have a low CV between the range of ~0.23 to ~0.85% which means very low variation regardless of magnitude.

b) Low Resistivity Validation with Aluminium Foil

Low resistivity measurement is validated with using UK supermarket aluminium foil with an estimated thickness of 12 microns. The aluminium foil sample is prepared in two ways, one for the TLM measurement is a 1-meter x 5mm long strip the other for 4-point probe measurement is a 10cm square.

- TLM Measurement

For the TLM Measurement, the function for resistance was determined using two instruments for our baseline data, and plotting the resistance measurement across 20 data points from 0 to 100 cm (5cm intervals). From determining its function of resistance in a slope-intercept, we can derive their measured sheet resistance values and measured bulk resistivities. The results are presented in Table 2, where we compare them with 4PP results and approximate resistivity for pure aluminium at ~2.65 $\mu\Omega\,\boldsymbol{cm}$ [12].

- 4PP Measurement using DIY 4 Point Probe

For the 4PP measurement, the DIY 4 Point probe was used to create a 10x10 Spatial map (100 data points) across a 10x10cm sheet of aluminium. Sheet resistivity was calculated using **Eq.3** and the bulk resistivity using **Eq.4**. The measured results are compared to values from TLM Measurement, and a reference value for aluminium, presented in **Table 4.** The statistics of the Aluminium measurements confirm our measurement of the aluminium sample has high precision with 99% of the points within the 3-sigma range, and 76% inside the one sigma range. In addition, the system successfully identified resistivity with 11.37% difference to a target reference value for aluminium. Therefore, the system successfully characterised Aluminium and performs sufficiently well in low resistivity measurements.

c) High Resistivity Validation with Graphite

High resistivity measurement is observed and validated with several samples of 6B pencil graphite on paper. Despite graphite being allotrope of carbon, a conductive material, graphite deposit on paper (pencil trace) has high resistivity due to it being impure layer of microscopic flakes [16]. Pencil graphite's resistivity also varies by composition and grades. Previous research had documented the resistivity for several common grades of pencil graphite including 6B [17].

We assume the deposits to be ~6 microns thick based on estimated surface roughness of copy paper (~5 microns) [18]. The graphite samples are prepared in three ways,

1. filling a 10cm square sheet of paper with 6B pencil
2. lightly erasing the left half of the same sheet, to reduce the graphite content and expose the paper
3. a smiley face

Sample 1 gives us the bulk resistivity distribution across a full graphite sheet, while samples 2 and 3 serve to give an illustration of sheet resistance variation and contrast between graphite and paper. Results are presented in **Table 5**.

The statistics confirm that our measurement has high precision with 100% of the points within the 3-sigma range, and ~65% inside the 1-sigma range. The calculated resistivity matches a reference value for Light pressed 6B Pencil Graphite closely (11.34% difference). Therefore, the successfully characterised "Light" 6B Graphite and performs sufficiently well in medium-high resistivity measurements

.

d) GST Wafer Repeatability Validation

Before measuring the GST wafers, validation is needed to estimate how much inconsistency is there between each reading. We executed three consecutive, spatial scans on the GST wafer across three distinct thermal states. First, we created three spatial maps the Amorphous state at approximately ~25℃, then using a hot plate, we crystallised the wafer to reach a nominal 200℃ and 300℃ and created the spatial maps for their sheet resistivities.

Since we are comparing only three data points for each of the 177 points, there is insufficient data for each point. Instead, we will compare their distribution of CV% across the three spatial maps, to map the spread of measurement fluctuation around the mean of the 177 points. The statistics of the spatial CV% are presented in **Table 6**, with

- Amorphous (25°C): Mean CV% = 55.11%, σ = 33.79%.
- Crystalline (200°C): Mean CV% = 39.12%, σ = 35.64%.
- Crystalline (300°C): Mean CV% = 30.13%,σ = 37.38%.

The System has a high mean CV, meaning that the next measurement would differ at an average of 30~55%. So, it could not be considered sufficiently precise and more data is needed. However, there is a progressive decrease in the mean CV% upon crystallisation (from 55.11% to 30.13%), as the sample transitions from the high-resistivity amorphous phase to the low-resistivity crystalline phase. This suggests that the precision improves on lower resistivity measurements.

Precision Resistor Validation

Rated (Ω)	*DMM (Ω)*	*N*	*μ (Ω)*	*Diff (%)*	*σ (Ω)*	*3σ%*	*2σ %*	*1σ %*	*CV%*
$\mathbf{10 \pm 0.1\%}$	9.91	10	9.983	0.737	0.0228	100	100	60	**0.2284**
$\mathbf{100 \pm 0.1\%}$	99.89	10	99.46	0.430	0.2367	100	100	60	**0.2380**
$\mathbf{1K \pm 0.1\%}$	985.7	10	1026	4.088	8.710	100	100	80	**0.8488**
$\mathbf{10K \pm 0.1\%}$	9746	10	9829	0.852	1.813	100	100	70	**0.1845**

Table 3. Precision Resistor Validation

Aluminium Resistivity Validation

Measurement	*μ (Ω cm)*	*Diff (%)*	*N*	*σ(Ω cm)*	*3σ%*	*2σ %*	*1σ %*	*CV%*
4PP (this paper)	2.99×10^{-6}	-	100	2.03×10^{-7}	99	96	76	**6.802**
Keithley 385	3.51×10^{-6}	-17.39	-	-	-	-	-	-
Rapid 318	2.70×10^{-6}	+9.70	-	-	-	-	-	-
Reference[12]	2.65×10^{-6}	+11.37	-	-	-	-	-	-

Table 4. Low Resistivity Validation using Aluminium

Graphite Resistivity Validation

Measurement	*μ (Ω cm)*	**Diff (%)**	**N**	**σ(Ω cm)**	**3σ%**	**2σ %**	**1σ %**	**CV%**
4PP 6B (this paper)	1.305	-	96*	0.7503	100	96.88	64.58	**57.47**
6B: Light Press [17]	1.157	+11.34	-	-	-	-	-	-
6B: Medium Press [17]	0.224	+82.86	-	-	-	-	-	-
6B: Hard Press [17]	0.106	+91.88	-	-	-	-	-	-

**Note = 4 points (bright spot outliers in the spatial map) were excluded. See Appendix*

Table 5. High Resistivity Validation using Graphite

GST Film Repeatability Validation

State	*μ (CV%)*	**N**	*σ*(CV%)
Amorphous	55.11	177	33.79
Crystalline 200	39.12	177	35.64
Crystalline 300	30.13	177	37.38

Table 6. GST Film Repeatability Validation

2. *Optical Reflectance*

To validate the optical reflectance measurement subsystem, the system's performance was evaluated using three common wavelengths critical to optical data storage, a common application of the GST Phase Change Material [19], 405nm (Blu-ray), 640nm (DVD), and 855nm (NIR). We perform a 15x15 spatial grid scan over a pencil graphite smiley face pattern on standard white copy paper across the aforementioned three wavelengths

Cluster	N	Channel	Mean	σ	CV%
Dark	23	F1_405nm	17.18	7.4	43.07%
		F6_640nm	28.56	7.24	25.36%
		NIR_855nm	18.14	7.35	40.52%
Intermediary	35	F1_405nm	52.72	11.08	21.02%
		F6_640nm	60.56	9.28	15.32%
		NIR_855nm	53.72	11	20.47%
Light	42	F1_405nm	85.56	6.46	7.55%
		F6_640nm	89.09	5.44	6.10%
		NIR_855nm	86.21	6.33	7.35%

Table 7. Reflectance Validation on Graphite Pattern

From **Eq.10,** We've calibrated white paper as the $I_{reference}(\lambda)$ or the reference intensity at max reflectance, and a completely dark condition (shroud, sensor pointing at black cloth) as $I_{dark}(\lambda)$. The distance between the light source/sensor and the target is 1.5cm, and the LED light source was driven with 12mA current.

The results are clustered into Dark, Intermediary, and Light using K-Means algorithm to find distribution of each cluster, presented in **Table 7.** The result provides a clear binary threshold to test the spectrometer reflectance measurement. The system cleanly distinguishes the threshold between light and dark regions across all three wavelengths.

VI. RESULTS AND DISCUSSION

This chapter presents the results of electrical (resistivity) and optical (reflectance) mapping and measurement to characterise a Phase Change Alloy, in which we will be targeting to characterise GST225.

The GST Film sample (S705) used in this paper came alongside provided sputtering/ deposition data that describes the material atomic concentrations and varying thicknesses across the wafer. It will be used to calculate the Bulk Resistivity and also to pinpoint the certain position on the wafer where there is the GST225 alloy.

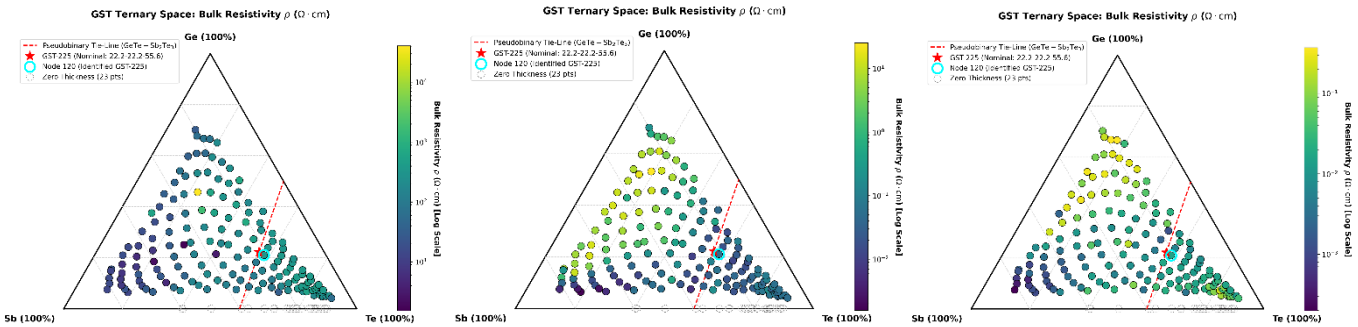
1. *Combinatorial Ge-Sb-Te Phase Space Mapping*

To map the physical properties across the continuous composition gradient of the GST wafer, we synchronized the spatial coordinates with the atomic concentration and thickness data. **Fig 8-9** displays the resistivity and reflectance ternary plots across the three different states of the wafer.

The bulk resistivity ternary maps **(Fig. 8)** capture a clear distinction between the three atomic components' intrinsic resistivities, and also the effect of crystallisation to the characteristics of the overall wafer.

In the amorphous state, the entire composition space is mostly highly resistive ($\rho = 10^{1}\sim10^{4}$ Ω·cm). Annealing to 200°C initiates crystallization, and the entire compositions resistivity drops to $10^{-2}\sim10^{1}$ Ω·cm After 300°C annealing, the entire composition drops to below 10^{-2} Ω·cm, a sign that it is transitioning to a highly conductive state.

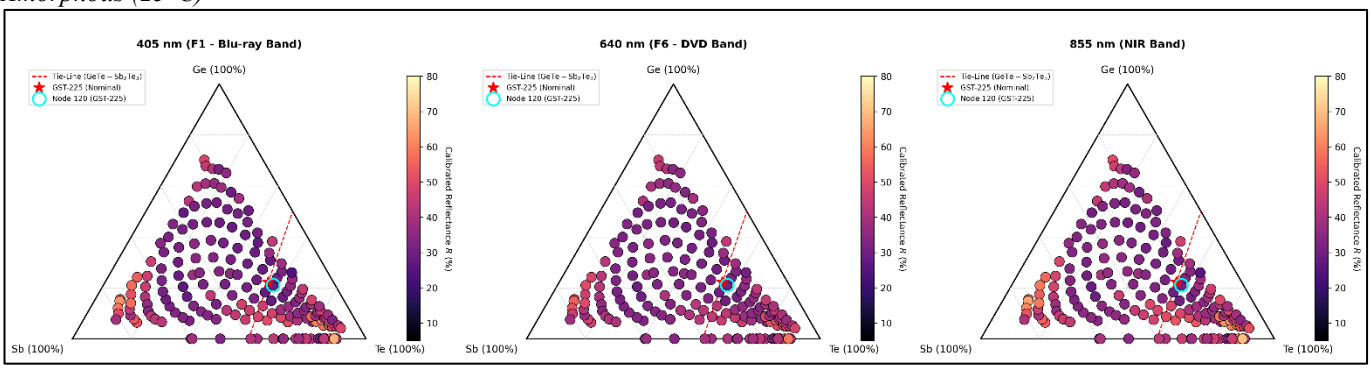
The reflectance ternary maps **(Fig. 9)** reveal a that Crystallization increases the reflectance percentage across all visible and NIR bands, showing a high reflectance contrast that aligns with the electrical characteristic becoming less resistive.



(a) (b) (c)

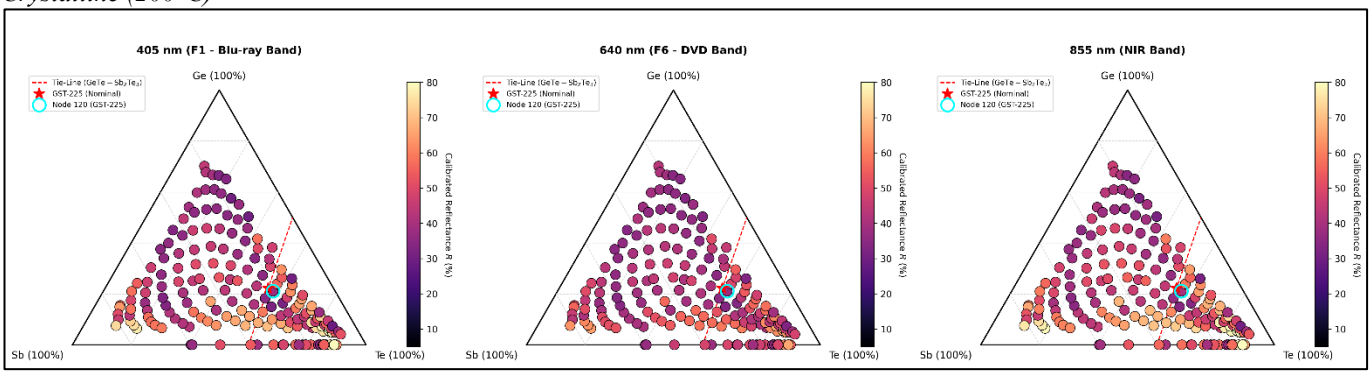
Fig 8. Bulk Resistivity Ternary Map of GST Sample at: (a) Amorphous state (25°C), (b) Crystalline, annealed to nominal 200°C, (c) Crystalline annealed to nominal 300°C

Amorphous (25°C)



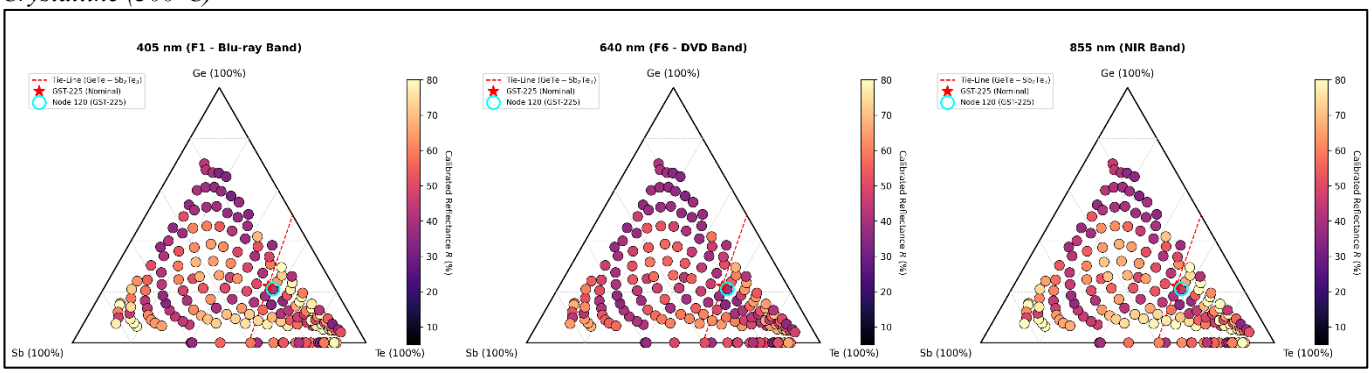
(a)

Crystalline (200°C)



(b)

Crystalline (300°C)



(c)

Fig 9. Optical Reflectance Ternary Map of GST Sample at: (a) Amorphous state (25°C), (b) Crystalline, annealed to nominal 200°C, (c) Crystalline annealed to nominal 300°C

2. Characterising the GST-225

Based on the atomic concentration data, we can approximate the location of the wafer to find the composition of GST225 at isolate Node 120, located at coordinates X = -132 mm, Y = -110 mm. We will observe the electrical (resistivity) and optical (reflectance) of the located node across its state transitions from Amorphous to crystalline. **Table 8** details the measured bulk resistivity for the GST255 node, alongside reference resistivity values for sputtered GST225 at its Amorphous state, Crystalline at 200, and Crystalline at 300 [20], while **Table 9** details the measured reflectance value for wavelengths at 405nm, 640nm and 855 nm compared to reference intrinsic reflectivity values from GST225's Refractive Index [21].

State	Measured ρ (Ω·cm)	Literature Value [20](Ω·cm)	Diff (%)
Amorphous (25°C)	3.315×10^{2}	2.8×10^{2}	0.1839
Crystalline (200°C)	5.480×10^{-2}	6.4×10^{-2}	−14.38%
Crystalline (300°C)	1.540×10^{-2}	1.44×10^{-2}	0.0694

Table 8. Node 120 (GST-225) Bulk Resistivity Measurement

Spectral Channel	Amorphous Intrinsic R (n-based) [21]	Crystalline Intrinsic R (n-based)[20]	Amorphous (25°C)	Crystalline (200°C)	Crystalline (300°C)	Lit. Contrast (ΔR)	Measured Contrast (ΔR)
405 nm (Blu-ray)	22.65%	12.23%	31.20%	40.50%	46.80%	13.50%	15.60%
640 nm (DVD)	36.62%	35.17%	34.80%	44.10%	50.40%	14.00%	15.60%
855 nm (NIR)	39.93%	46.44%	38.40%	48.90%	55.20%	17.00%	16.80%

Table 9. Node 120 (GST-225) Optical Reflectance Measurement

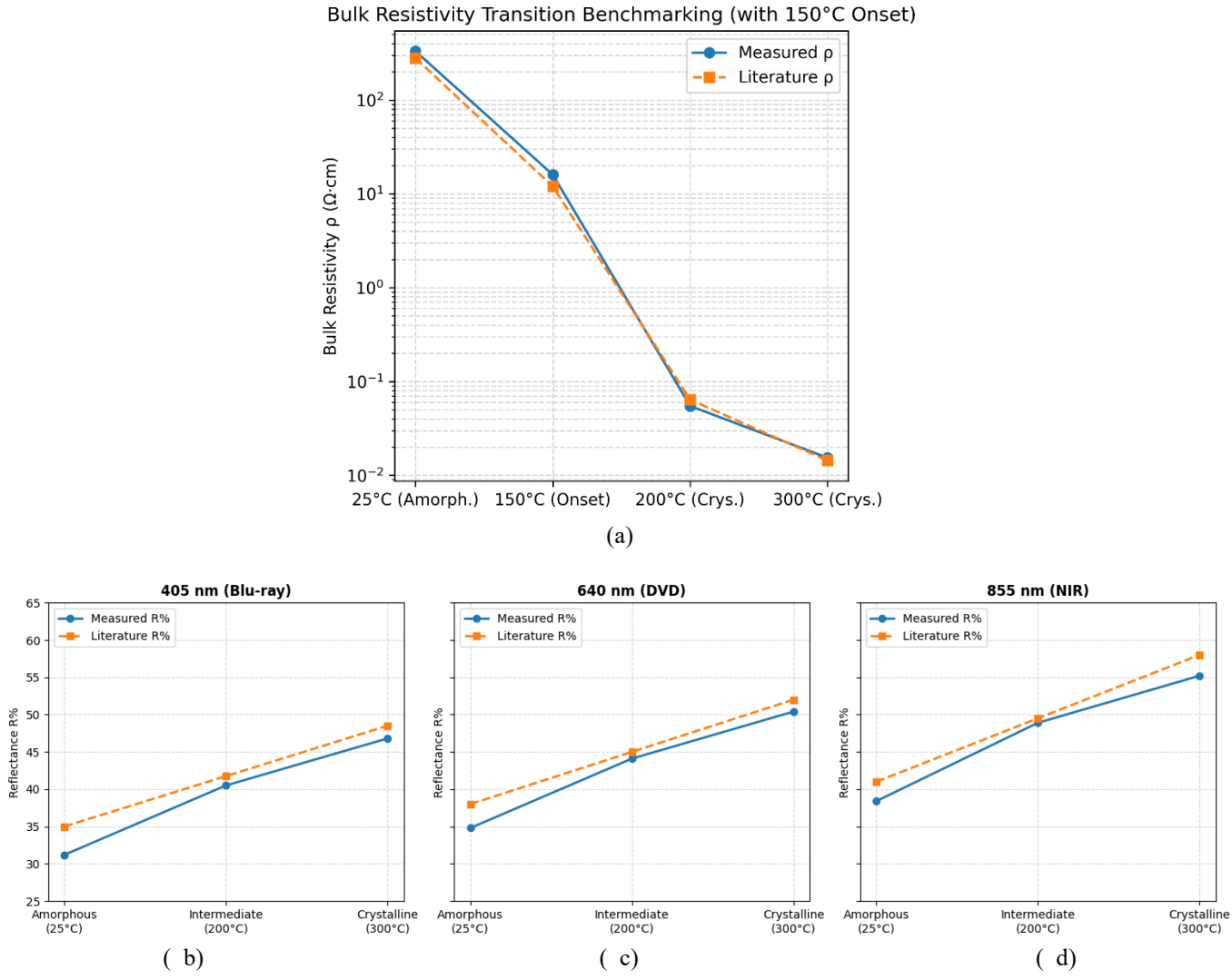


Fig 10. Phase transition plots for GST225 Sample for: (a) Bulk Resistivity (b) Reflectance at 405nm (c) Reflectance at 640 nm (d) Reflectance at 855 nm

As presented in **Table 8**, and **Fig 10 (a)** the GST225 measured at Node 120 aligns closely with established literature values across both its amorphous state and its crystalline phases. the high degree of correspondence demonstrates strong measurement accuracy, validating that the resistivity characterization system works as intended.

Similarly, the optical reflectance data at Node 120 in **Table 9** and Fig 10 (b-d) exhibits high degree of correspondence with the reference benchmarks. The

reflectance behaviour across all evaluated wavelengths reproduces the expected contrast profile observed in the literature, confirming that the optical reflectance setup performs reliably and yields consistent results.

VII. CONCLUSION AND FUTURE WORK

- Conclusion

This paper successfully demonstrates the development of a low-cost, modular, and autonomous multi-property spatial mapping system for CSAF developed on a repurposed CNC3018 mechanical platform. By integrating a 4-point probe measurement system for resistivity measurement and a 14-channel AS7343 spectrometer for optical reflectance measurement, the system successfully characterises both electrical and optical properties of a thin film sample.

The system was validated using precision resistors Aluminium foil and graphite deposit on paper, which all show sufficient accuracy and precision. When measuring Ge-Sb-Te CSAF wafer, we are able to identify ternary phase space mapping across three phase states (Amorphous, Crystalline at 200°C, and Crystalline at 300°C) to identify the precise boundary to get the composition for GST-225 alloy.

The system is able to achieves accuracy close to references and past documentation of the GST 225 and other sample materials prepared, and has sufficiently high throughput, as a 100mm wafer with 177 points is able scanned in under 30 minutes. With simple and affordable components, it has high reproducibility, becoming a potentially highly accessible tool for materials research.

- Challenges

Throughout the system's development, validation, and wafer mapping, we had identified a few challenges:

1. **Probe Contact Consistency**: Because thin-film layers on CSAF are highly delicate, repeated mechanical contact cycles is suspected to cause surface damage and this causes lowered repeatability, as evident by the high CV% in Table 6.

2. **The need for Dual Mode (Low Resistivity and High Resistivity modes)**: Because the PicoAmmeter can only handle 2mA maximum, a low resistivity sample such as Aluminium Foil has to be measured bypassing the PicoAmmeter to be able to feed higher current.

3. **Picoammeter Analogue Output:** The Keithley 485 Picoammeter supports digital output through the IEEE GPIB Protocol and connector. Due to complexity in interfacing such cable to a standard modern computer, we decided to go with the Analogue output of the Picoammeter (-1 V to +1V) which had to be flipped in polarity via OpAmp before going to the ADC Input that only accepts positive voltage.

- Future Work

To further expand the scope, precision and accuracy of this project, there are recommended development steps:

1. **Digital Switch for Dual Mode**: Instead of a manual DIP Switch that has to be turned on or off according to the mode that we wish to select, a digital switch controlled by the Arduino/MCU based on the command for telemetry would prevent accidents or mistakes in such cases where we accidentally push a high current into the PicoAmmeter.

2. **Utilising a SMU (Source Measurement Unit) in place of a Picoammeter + PSU**: using a high-resolution SMU would replace the need of both the Picoammeter and the Power Supply Unit since the SMU serves as the Current Source and can measure its own output. Modern SMUs with digital out can also interface directly with either the Arduino/MCU or PC.

ACKNOWLEDGEMENTS

It is in my utmost gratitude to say a warm thank you to my Supervisor, Dr Robert Simpson who has made this work possible. Thank you to the faculty members of Department of Electronic, Electrical and Systems Engineering of University of Birmingham for facilitating me throughout the process of working on this project, thank you to Indonesia Endowment Fund for Education Agency for granting me the scholarship to be able to pursue my MSc in the United Kingdom, and last but not least thank you to my closest family, friends and peers who have supported me along the way.

REFERENCES

[1] D. Priyadarshini, P. Kondratyuk, J. B. Miller, and A. J. Gellman, 'Compact tool for deposition of composition spread alloy films', *J. Vac. Sci. Technol. Vac. Surf. Films*, vol. 30, no. 1, p. 011503, Jan. 2012, doi: 10.1116/1.3664078.

[2] C.-Y. Lee *et al.*, 'Real-time Multi-instrument Autonomous Discovery of Novel Phase-change Memory Materials', 2026, *arXiv*. doi: 10.48550/ARXIV.2605.18033.

[3] P. Guo, A. M. Sarangan, and I. Agha, 'A Review of Germanium-Antimony-Telluride Phase Change Materials for Non-Volatile Memories and Optical Modulators', *Appl. Sci.*, vol. 9, no. 3, p. 530, Feb. 2019, doi: 10.3390/app9030530.

[4] P. I. Lazarenko *et al.*, 'Electrical properties of the Ge2Sb2Te5 thin films for phase change memory application', presented at the REVIEW OF PROGRESS IN QUANTITATIVE NONDESTRUCTIVE EVALUATION: Proceedings of the 35th Annual Review of Progress in Quantitative Nondestructive Evaluation, Chicago (Illionois), 2016, p. 020013. doi: 10.1063/1.4945968.

[5] Moh. Toifur, Moh. I. Sukarelawan, and O. Okimustava, 'Home Made Four-Point Probe: Case Studies of the Wobbly A and B Probes', *Indones. Rev. Phys.*, vol. 6, no. 1, pp. 56–62, Jun. 2023, doi: 10.12928/irip.v6i1.8362.

[6] H. Chandra, S. Allen, S. Oberloier, N. Bihari, J. Gwamuri, and J. Pearce, 'Open-Source Automated Mapping Four-Point Probe', *Materials*, vol. 10, no. 2, p. 110, Jan. 2017, doi: 10.3390/ma10020110.

[7] F. M. Smits, 'Measurement of Sheet Resistivities with the Four-Point Probe', *Bell Syst. Tech. J.*, vol. 37, no. 3, pp. 711–718, May 1958, doi: 10.1002/j.1538-7305.1958.tb03883.x.

[8] G. Tuttle, ‘Contact resistance and TLM measurements’. Dept. of Electrical and Computer Engineering, Iowa State University.
[9] E. Hecht, *Optics*, 5 ed/fifth edition, Global edition. Boston Columbus Indianapolis New York San Francisco Amsterdam Cape Town Dubai London Madrid Milan Munich: Pearson, 2017.
[10] I. Ohlídal, J. Vohánka, J. Dvořák, V. Buršíková, and P. Klapetek, ‘Determination of Optical and Structural Parameters of Thin Films with Differently Rough Boundaries’, *Coatings*, vol. 14, no. 11, p. 1439, Nov. 2024, doi: 10.3390/coatings14111439.
[11] M. Mejia-Herrera, J. S. Botero-Valencia, J. F. Vargas-Bonilla, and J. M. Pearce, ‘An open-source spectral measurement platform for plant reflectance and material identification’, *HardwareX*, vol. 26, p. e00768, Jun. 2026, doi: 10.1016/j.ohx.2026.e00768.
[12] V. Polyakov, ‘Resistivity of Aluminum’. 2004. Accessed: Aug. 20, 2026. [Online]. Available: https://hypertextbook.com/facts/2004/ValPolyakov.shtml
[13] D. G. Altman and J. M. Bland, ‘Standard deviations and standard errors’, *BMJ*, vol. 331, no. 7521, p. 903, Oct. 2005, doi: 10.1136/bmj.331.7521.903.
[14] W. Zhang, J. Xu, S. Wang, Y. Zhou, and J. Mi, ‘Metrology Challenges in 3D NAND Flash Technical Development and Manufacturing’, *J. Microelectron. Manuf.*, vol. 3, no. 1, pp. 1–8, 2019, doi: 10.33079/jomm.20030102.
[15] E. Theodorsson, ‘Understanding and expressing variation in measurement results’, *J. Lab. Precis. Med.*, vol. 10, no. 1, pp. 3–3, Jan. 2025, doi: 10.21037/jlpm-24-53.
[16] A. Shukla, A. Gaur, V. S. Chauhan, and R. Vaish, ‘A Review on Pencil Drawn Thermoelectric Devices on Paper’, *ECS Sens. Plus*, vol. 3, no. 2, p. 026401, Jun. 2024, doi: 10.1149/2754-2726/ad53dd.
[17] C. Inyeo and P. Wattanakasiwich, ‘Pencil on paper as low cost alternative resistors’, *J. Phys. Conf. Ser.*, vol. 1144, p. 012165, Dec. 2018, doi: 10.1088/1742-6596/1144/1/012165.
[18] M. Santhiago, M. Strauss, M. P. Pereira, A. S. Chagas, and C. C. B. Bufon, ‘Direct Drawing Method of Graphite onto Paper for High-Performance Flexible Electrochemical Sensors’, *ACS Appl. Mater. Interfaces*, vol. 9, no. 13, pp. 11959–11966, Apr. 2017, doi: 10.1021/acsami.6b15646.
[19] D. Sahoo and R. Naik, ‘GSST phase change materials and its utilization in optoelectronic devices: A review’, *Mater. Res. Bull.*, vol. 148, p. 111679, Apr. 2022, doi: 10.1016/j.materresbull.2021.111679.
[20] I. Friedrich, V. Weidenhof, W. Njoroge, P. Franz, and M. Wuttig, ‘Structural transformations of Ge2Sb2Te5 films studied by electrical resistance measurements’, *J. Appl. Phys.*, vol. 87, no. 9, pp. 4130–4134, May 2000, doi: 10.1063/1.373041.
[21] J. A. Frantz *et al.*, ‘Optical constants of germanium antimony telluride (GST) in amorphous, crystalline, and intermediate states’, *Opt. Mater. Express*, vol. 13, no. 12, p. 3631, Dec. 2023, doi: 10.1364/OME.506019.

APPENDICES

APPENDIX 1. USAGE OF GENERATIVE AI

In accordance with university academic integrity and assessment guidelines,
I declare that Generative Artificial Intelligence (AI) was utilized in the preparation of this dissertation and does not reflect the final work. It is used with details as below:

1) Google Notebook LM

- **Scope of Use:**
  - Generating Rough Drafts and Mind Maps based on PDFs and other materials uploaded.
  - Proofreading draft version of the paper

  Example Prompt: Generate a mind map for Chapter V. Design Validation based on the notes provided.
  Example Output:



2) Google Gemini (Large Language Model)

- **Scope of Use:**
  - Natural Language Processing/ Transcription to convert audio recordings during supervision meeting and spoken notes into written format or table.
  - Proofreading and help with polishing to improving clarity
  - AI Assisted Coding for Google Colab and Python

Example Prompt:

"Can you summarise the meeting recording on 27th of July into Bullet Points?

Example Result:

Here is a detailed, academically structured summary of your progress meeting with Dr. Simpson on **27 July 2026**:

🔦 **1. Four-Point Probe (4PP) & TLM Validation**

- **The Progress:** You presented your validation progress of the 4PP setup using the **Transmission Line Method (TLM)** on highly conductive aluminum foil to successfully isolate sheet resistance.
- **Accuracy Check:** By comparing your DIY setup readings with a commercial multimeter and a high-precision Keithley digital multimeter, you demonstrated a **96% measurement accuracy** on low-resistance aluminum samples.
- **Academic Directives:** Dr. Simpson praised this validation work and strongly advised that you explicitly include this empirical data and statistical metrology in your report to secure high marks.

📸 **2. Optical Reflectance Sensor Geometry & Calibration**

- **Specular vs. Diffuse Reflection:** Dr. Simpson corrected your assumption that specular reflection is "bad data" that prevents color separation, explaining that the spectrometer detector intrinsically splits white light into color channel intensities.
- **Optical Geometry Optimization:** He advised against steep illumination angles, suggesting instead that the detector breakout board be mounted close to **normal incidence (or at 45°)**.
- **Z-Axis Height & Spot Size:** You discussed changing the sensor height relative to the sample. Because the compositional grid squares on the thin-film wafer are extremely small (~5 mm), the Z-axis height must be kept very close—approximately **2 to 2.5 mm (0.1 inches)**—to prevent the optical spot size from overlapping adjacent grid squares.
- **Height Calibration Strategy:** Dr. Simpson noted that as the sensor gets closer to the sample, the light intensity drops but the spatial resolution improves. He suggested taking scans at multiple heights; you should see the spectral signature oscillate clearly as the carriage passes from bare silicon substrate to the GST alloy, confirming optimal resolution.

⚠️ **3. Sample Protection & Probing Sequence (Crucial Safety Rules)**

- **Protecting the Jandel Probe:** Dr. Simpson expressed serious concern about protecting his expensive physical 4PP head, stressing that you must be completely confident in your G-code limits to avoid blunting the needles or crashing the carriage.
- **Order of Measurements:** He issued a strict directive that you must perform all **non-contact optical spectral measurements *before* executing physical 4PP electrical probing**. This is because the physical contact of the gold-plated pogo pins leaves visible mechanical indents and scratches on the fragile sample, which would permanently spoil subsequent optical reflectance mappings.

🗓️ **4. Supervisor Availability & Logistics**

- Dr. Simpson arranged to hand over the physical samples immediately after the meeting.
- He noted that he would be away for several conferences for most of August, making it critical to complete any physical or hardware assistance requests immediately.

APPENDIX 2. CODE REPOSITORY

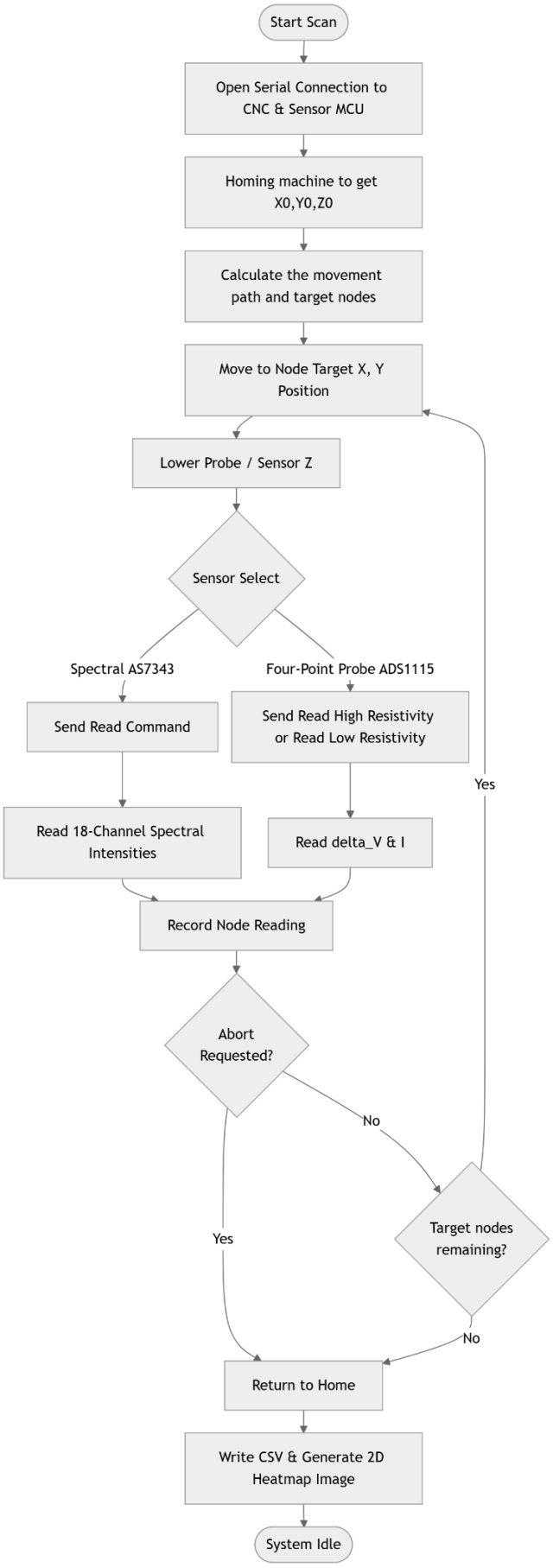
The code repository including the Python code, the Arduino Nano telemetry INO files, and the exploratory data analysis notebooks including all data plots in this paper is maintained at:

https://github.com/irfanbstr/MScFinalProject

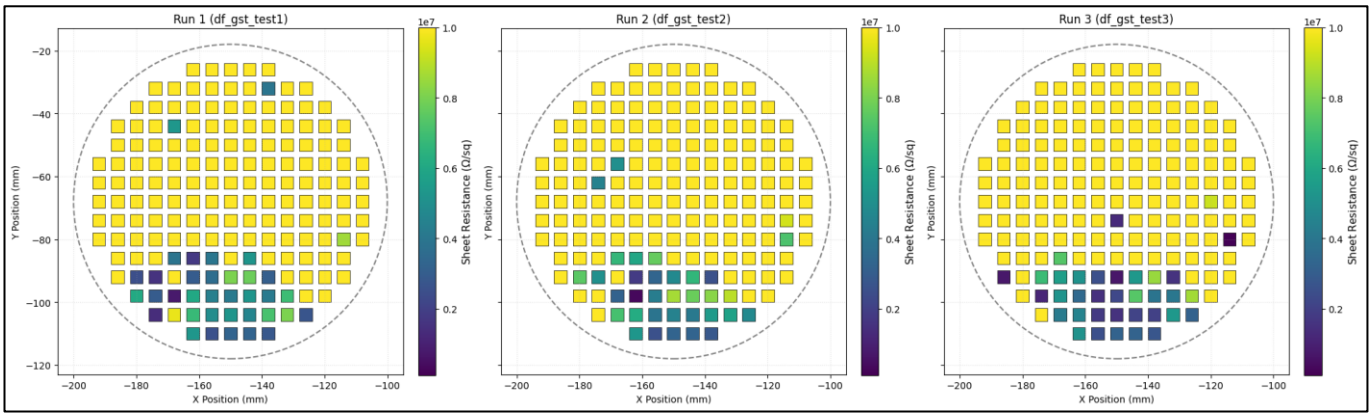
The Complete Exploratory Data Analysis is available as a Google Colab Notebook at:

https://colab.research.google.com/drive/1kaqdY1cxI8NlREmFfgUynw80iRmRbIqG?usp=sharing

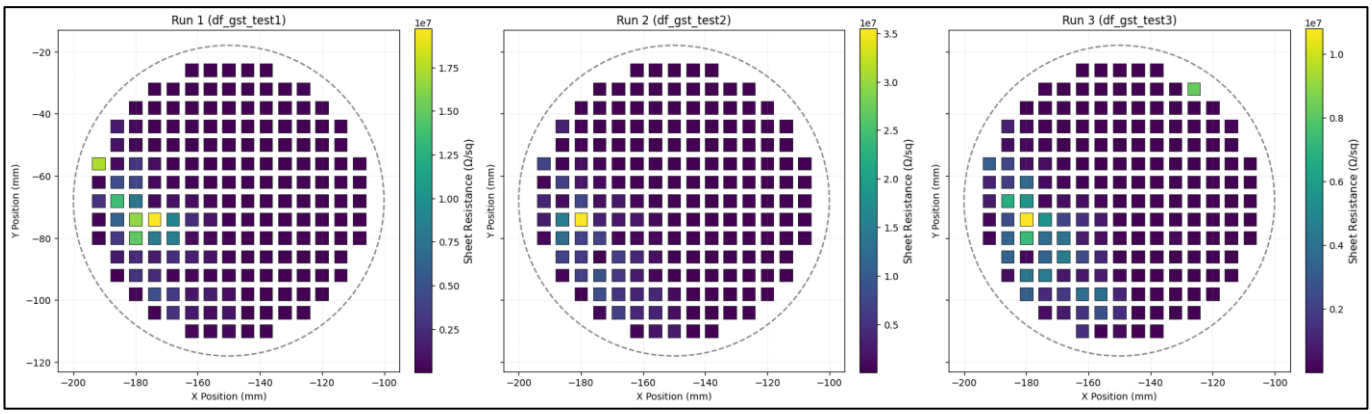
APPENDIX 3. COMPLETE SYSTEM SOFTWARE FLOWCHART



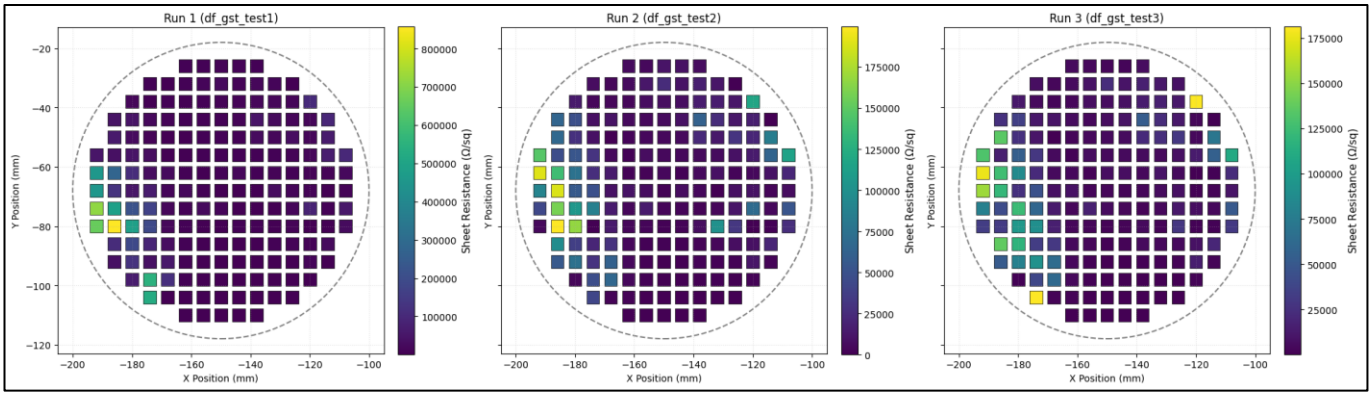
APPENDIX 4. WAFER SPATIAL MAPPING SCANS



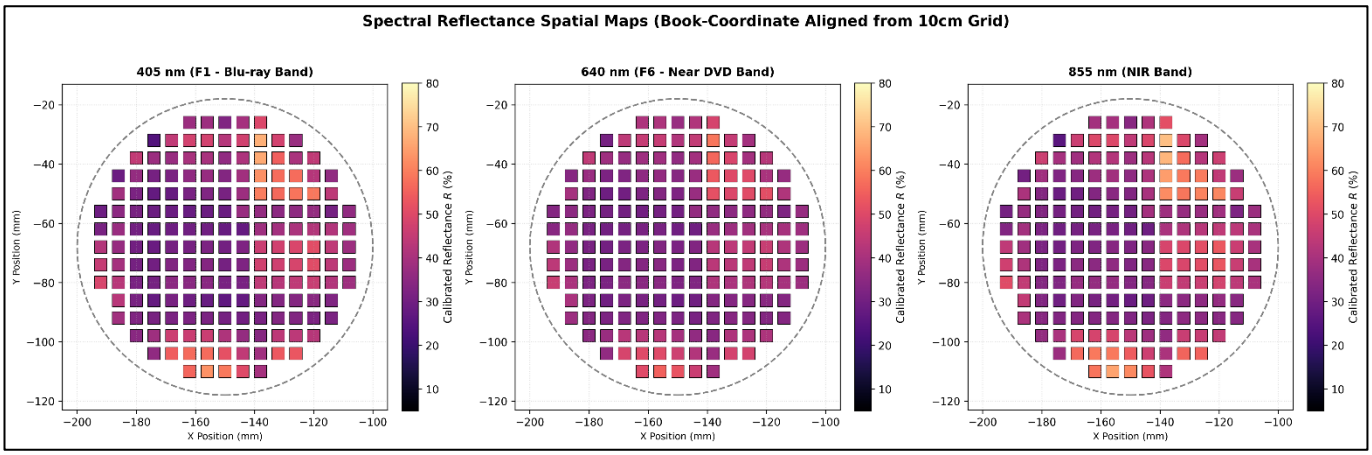
Sheet Resistivity Spatial Scan of 3 different Runs of GST sample (Amorphous)



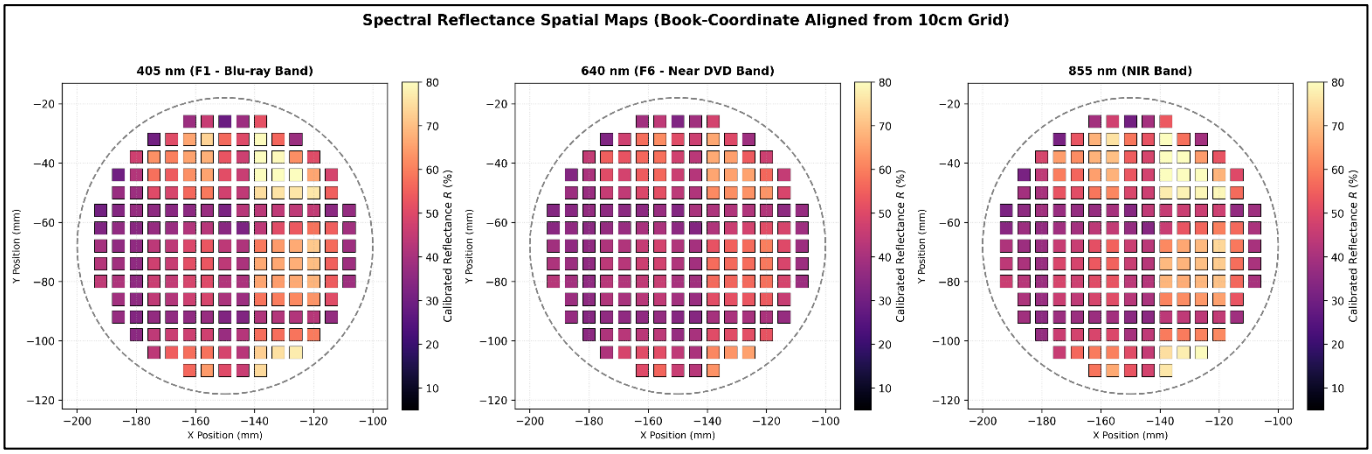
Sheet Resistivity Spatial Scan of 3 different Runs of GST sample (Crystalline 200°C))



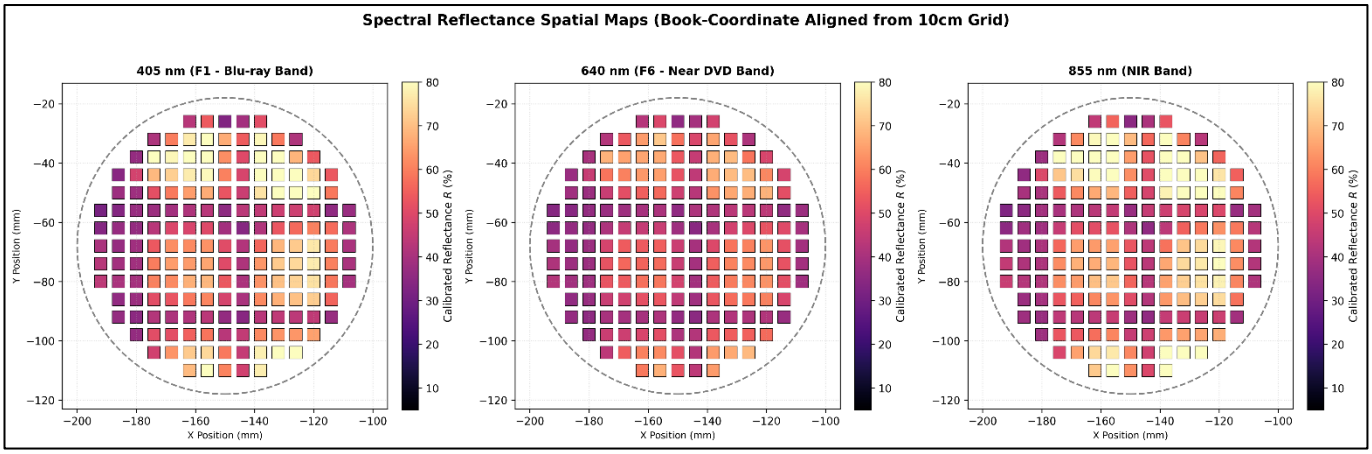
Sheet Resistivity Spatial Scan of 3 different Runs of GST sample (Crystalline 300°C))



Reflectance Scan of GSTS075 Sample (Amorphous) at 405nm (left), 640nm (centre), and 855nm(right)



Reflectance Scan of GSTS075 Sample (Crystalline 200°C) at 405nm (left), 640nm (centre), and 855nm(right)



Reflectance Scan of GSTS075 Sample (Crystalline 300°C) at 405nm (left), 640nm (centre), and 855nm(right)

More Data Plots on Google Colab:

https://colab.research.google.com/drive/1kaqdY1cxI8NlREmFfgUynw80iRmRbIqG?usp=sharing